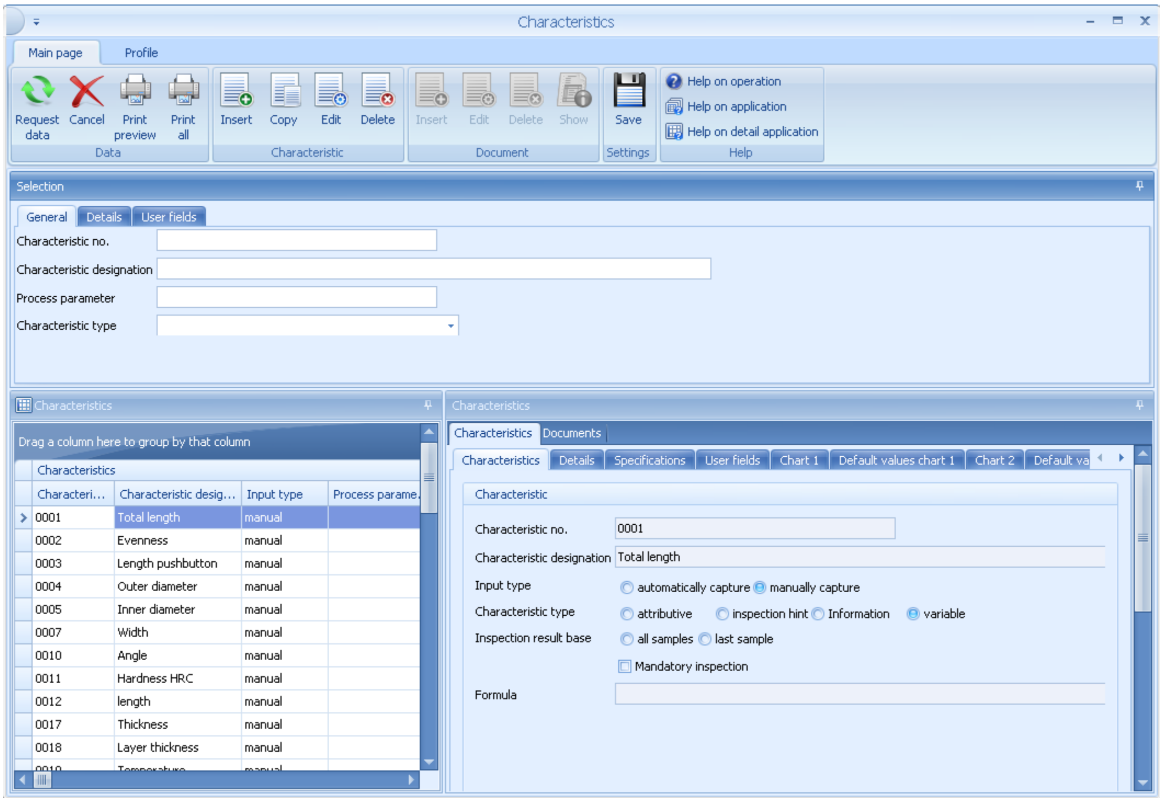
1 Characteristic Master Data

Overview

Menu	Master data → Quality management → Characteristics
Transaction code	chrq
Function authorization	chrq

Available user fields		
Where?	**Object type/user field key**	**Source (type)**
Table and detail view	CMM/SYSTEM	QM
How to configure user fields?	Which user field types are available?	

Purpose



The catalog of characteristics has been designed to define characteristics and, as a result, to predefine characteristic data of inspection plans. For this reason, it aims at people involved in inspection planning.

The catalog of characteristics is one of the most important basic catalogs. You cannot set up any inspection plan without this catalog. As this catalogue is used to predefine characteristics for inspection plans, it includes extensive input options. Basically, the catalog of characteristics should only include such data, which will not have to be modified when the characteristics are assigned to the inspection plan later on. For example, the definition of limit values is usually not reasonable as these values are only known when an inspection plan is set up. Only when you assign data to an inspection plan, a relation between data and article is established. Note this and you will know what kind of information you should predefine. For example, it must be carefully considered whether the characteristic "outer diameter" is only created once and detailed information is stored in the inspection planning later on or whether several "outer diameter characteristics" are created, e. g. with specification of limit values. Usually, it is an advantage to store a restricted number of general characteristics. The required evaluations/reports also play a role in this context. If a new "outer diameter characteristic" is created for almost every tolerance change, this characteristic is "valid" for one article only. In a subsequent failure analysis, a comprehensive evaluation is not possible in this case!

It is important that each detail defined here can be modified in the inspection planing later on or that details, which have not been stated, can still be added.

The configurations made in the characteristics' master data are not final. The characteristics' master data is used as a template for later inspection planning. You can complete and modify all settings of the characteristics' master data during inspection planning.

Integration

The catalog of characteristics is a global catalog that is used in many QM applications. Please find below some possible fields of application that refer to the catalog of characteristics.

- Inspection planning for production, goods receipt, goods issue, initial samples and calibration
- Inspection requirements for production, goods receipt, goods issue, initial samples and calibration
- Failure analysis in complaint management
- Several reports/evaluations

Requirements

There are no special requirements.

Selection criteria

The application provides the following selection criteria:

- Characteristic no.:
  Number of the characteristic

- Characteristic designation:
  Designation of the characteristic – Note: You may use wildcards "*"

- Characteristic type:
  Inspection type: attributive, inspection chart, variable

Tab *Details*

- Gage
  Select a gage

- Gage designation:
  Select a gage designation

Tab *User fields*

- If user fields are created, they may be selected

If several selection criteria are used, overlapping results are displayed in the characteristics' master data.

In addition, the column filter allows for the content of each individual column to be filtered.

Field descriptions

The available fields are self-explanatory and are not explained separately, except for the address fields.

Tab *Characteristics*

Characteristic no.

Unique number of the characteristic

Characteristic designation/name

Designation of the characteristic

Input type

Automatic or manual data collection. This field controls the release of HYDRA-PDV fields (in case of automatic collection). If the automatic collection function is selected, the characteristic type is restricted to the "variable" option.

Characteristic type

This option specifies whether the collection of measured values (variable) or the identification of the number of detected failures (attributive) is used for the inspection. If you select the attributive inspection, use the input type to define whether the collection should be based on a catalog or whether the standard collection is performed. Further characteristic types are the inspection chart and the information characteristic. If you select the inspection chart, you can enable the input type *visual defects recording.* The information characteristic is only used to display a document during the inspection process. Subject to the input type, the lower area of the dialog provides the respective sampling schemes.

Visual recording: The characteristic document (not the *inspection requirement document*) is displayed with the position 1. This must be type FILE. The system supports these formats: JPEG, JPG, PNG. To divide a graphic in different areas, you must define the grid for the x-axis and the y-axis (e.g. A,B,C,D,E)

The catalog-based collection and the visual defects recording are only available, if the extension QMCharacteristicExtendedTyp is enabled. Both collection types are only available, if the collection is performed at a specified inspection point.

Inspection result base

This setting defines whether all samples or only the sample recorded last is used to identify the inspection result (pass/fail).

Mandatory inspection

If this option is activated, you must enter at least one measured value for this characteristic, before you can complete an inspection order including this characteristic.

Formula:

See chapter Calculation of formulas.

To display this field in the inspection plan characteristics, you require the authorization "iriscp.formula".

Tab Details

Group *Gage*

Gage

Defines whether a gage or gage group is to be assigend to the characteristic:

Assignment of the gage (or gage group) to be used.

You can also use resources of resource type "PRM" of the resource management. To display this field in the inspection plan characteristics, you require the authorization "iriscp.gage".

Gage designation (name of test equipment)

Shows the name of the gage

Group *Properties*

Certificate printing

The selected option defines whether this characteristic is to be printed (display selection or print always) or not (print never) when certificates are printed at a later stage (e.g. acceptance, inspection certificate). If you select the option "display selection", a list of the characteristics with this printing option set is displayed prior to printing. In the list, these characteristics are preselected for the print of a certificate. However, this selection may be removed. Finally, all selected characteristics and the characteristics with the "print always" option are included in the certificate print. Characteristics with the "print always" option do not appear in a selection list, as they are printed in any case. Please note that this option only affects certificate forms.

Failure weighting

If the inspection result for the characteristic is "fail", you can classify the result here for information purposes.

Group *Inspect*

Analyseauswahlkatalog

Here, you can select an analysis selection catalog. The catalog restricts the selection of possible failures you can enter (failure types, failure location, etc.). (All available failures may still be entered, if you directly enter their number).

Designation of analysis selection

Shows the designations of analysis selection catalogs

Tab *Specifications*

Once the "specifications" tab has been selected, the sample scheme and constructional measures may be entered. In this context, it has to be considered that (as already mentioned) the definition of tolerance limits in the master data of characteristics is only reasonable if certain conditions are met. The same applies to the definition or calculation of action and warning limits. This section explains the possibilities in detail.

Group *Sampling scheme*

Sampling scheme

The following sampling schemes are available:

- 100% inspection
- k value inspection
- lot inspection

- n-c inspection
- SPC inspection

The sampling scheme defines the inspection procedure. In case of an n-c inspection and parameters 5-0, 5 pieces are checked and 0 failures may be detected.

Find a more detailed description in section Sampling **schemes**.

Sample size/expected sample size

Specification of the sample size (number of samples) or the expected sample size depending on the sampling scheme, see section Sampling schemes.

Acceptance quantity

Acceptance quantity for the n-c inspection, please also see section Sampling schemes.

Interval type

Input for SPC or n-c inspections: time, pieces, once, none. See chapter Sampling schemes .

Interval value

Specifies the interval subject to the interval unit.

To display this field in the inspection plan characteristics, you require the authorization "iriscp.interval".

Interval unit

For n-c or SPC inspections, e.g. minutes, hours.

With output batch change

If the output batch changes, an inspection becomes due.

To display this field in the inspection plan characteristics, you require the authorization "iriscp.interval".

Note:

The option *With output batch change* only triggers the generation of an inspection point, if the respective change of the output batch is included in the dialog "Change of batches" (dialog ID: CA_WL). For example, reel cutting dialogs do not generate inspection points.

With machine status change

If the machine status changes, an inspection becomes due.
To display this field in the inspection plan characteristics, you require the authorization "iriscp.interval".

Source status

Here, you can specify source statuses (specific non-productive machine statuses) – separated by commas. If the machine then changes from a specified source status into a productive machine status, an inspection becomes due.

To display this field in the inspection plan characteristics, you require the authorization "iriscp.interval".

As of SP8, the following configurations are available in addition.

- The field is completely empty: For this characteristic, a machine status change always generates an inspection point, if the machine changes from a non-productive into a productive status.
- "x-y", comma-separated: If the machine changes from source status x to target status y (may be non-productive), an inspection point is generated for this characteristic.
- "x-": If the machine changes from source status x to an arbitrary target status (may be non-productive), an inspection point is generated.
- "-y": If the machine changes from an arbitrary source status to target status "y" (may be non-productive), an inspection point is generated.

With change of shifts

An inspection becomes due on changing shifts.
To display this field in the inspection plan characteristics, you require the authorization "iriscp.interval".

Inspection due date of last off inspection

For details on the configuration of a last off inspection, refer to the section "Last off inspection".

Group *Constructional measures*

Unit

Pieces, meter, kg, etc. Unit of the characteristic. Allocate the units by using the unit catalog.

Decimal places

Number of decimal places. Leading zeros before the comma are not displayed in the specification fields. By default, the number of decimal places defined in the system settings is pre-assigned.

Size (measure type)

Plausibility and tolerance limits can be entered as absolute, relative or percentage values. Please note that relative or percentage lower limits (lower tolerance limit, lower process limits) must be specified with a negative algebraic sign.

Standard

Calculation of tolerances based on specific standards (e.g. ISO metric fits). Subject to the selected standard, further information is requested (e.g. engineering fit). The system automatically calculates the tolerance limits on the basis of these specifications.

Fit

Calculation of tolerance limits on the basis of a specific standard and engineering fit. The selected fit depends on the selected standard.

Upper PL

Specfies the upper plausibility limit

Upper TL

Specifies the upper tolerance limit (upper specification limit)

Target value

Specifies the target value

Lower TL

Specifies the lower tolerance limit (lower specification limit)

Lower PL

Specifies the lower plausibility limit

Generate failure (UTL)/(LTL)

If measured values are recorded and the checkbox *Generate failure* is enabled, a violation of the limit value automatically results (in the background) in the failure type "limit value violation" (AUTO:TG> or AUTO:TG<). This option is not available for attributive characteristics, as the specification is only used for information purposes in this case.

User fields tab

If you have defined user fields for characteristics, they are displayed and may be edited here.

Tab *Chart 1/Chart 2*

In tab chart1/chart2, you can define the control charts to be used. These control charts are later available in the integrated measurement recording and in the measurement recording for terminals (SPCM). You can define a total of two different control charts. Here, you can store for each control chart the action limits, warning limits and the mean value of variable characteristics. There are two different possibilities to define these limit values. You can enter the limit values manually or the limit values are calculated using the specified default values included in tab *Default values chart1/2*. For further information on control charts, refer to sections 1.2 Control charts for variable characteristics and 1.3Control charts for attributive characteristics.

Chart 1 / Chart 2

Specifies the control chart displayed in the measurement recording dialog on the terminal. You can define action limits on the basis of the control chart type.

Upper AL

Specifies the upper action limit. The system can calculate the value using the default values, if the checkbox *Calculate* is enabled. (See also Control charts for variable characteristics).

Upper WL

Specifies the upper warning limit. The system can calculate the value using the default values, if the checkbox *Calculate* is enabled. (See also Control charts for variable characteristics).

MV (Mean value)

Specifies a mean value, e.g. as basis for the automatic calculation of limits by the system.

Lower WL

Specifies the lower warning limit. The system can calculate the value using the default values, if the checkbox *Calculate* is enabled. (See also Control charts for variable characteristics).

Lower AL

Specifies the lower action limit. The system can calculate the value using the default values, if the checkbox *Calculate* is enabled. (See also Control charts for variable characteristics).

Generate trend error

The option "generate trend error" has to be activated to be able to generate an automatic error if a trend exists (e.g. seven values in a row are descending or ascending, the number of values is defined while the system is customized). To identify a trend, the samples of an inspection step characteristic are checked, sorted by their sample number – regardless of the machine where the data has been recorded.

Generate error (UWL) / (LWL)

Enable the checkboxes *Generate error (UWL) / (LWL)* to generate automatically (in the background) the failure type "Limit value violation" (AUTO:WG> or AUTO:WG<), if a limit value is violated during the recording of measured values. Here, the violation of the limit value is identified using the stored control chart. In case an xq chart is stored, the automatic error is only generated if the respective xq value of the control chart, and not the single value, exceeds the warning limits.

Generate error (UAL) / (LAL)

Enable the checkboxes *Generate error (UAL)) / (LAL)* to generate automatically (in the background) the failure type "Limit value violation" (AUTO:EG> or AUTO:EG<), if a limit value is violated during the recording of measured values. Here, the violation of the limit value is identified using the stored control chart. In case an xq chart is stored, the automatic error is only generated if the respective xq value of the control chart, and not the single value, exceeds the action limits.

Tab *Default values chart 1 / Default values chart 2*

For further information on control charts, refer to sections Control charts for variable characteristics and Control charts for attributive characteristics.

Group *Default for calculating limit values*

Calculation type

Default values to calculate limit values: Cpk, Sigma, sq/an, Rq/dn, relative deviation from xq, deviation from xq in percent

Cpk

Default value of cpk

Sigma

Default value or calculated sigma value

Rq/sq (RQuer/sQuer)

Default value for Rq/sq (RQuer/sQuer)

Group *Non-action probability*

Action limits (non-action probability)

Specifies the action probability (only visible with the calculation type: cpk, Sigma, rQuer/sQuer)

Warning limits (non-action probability)

Specifies the action probability (only visible with the calculation type: cpk, Sigma, rQuer/sQuer)

Group *Deviation from xq specification*

rel. AL

Direct entry of the action limits (only visible with calculation types relative/percentage deviation).

rel. WL

Direct entry of the warning limits (only visible with calculation types relative/percentage deviation).

Group *Confidence interval*

Confidence interval

One-sided or two-sided. You can select one-sided or two-sided for the control charts R and s.

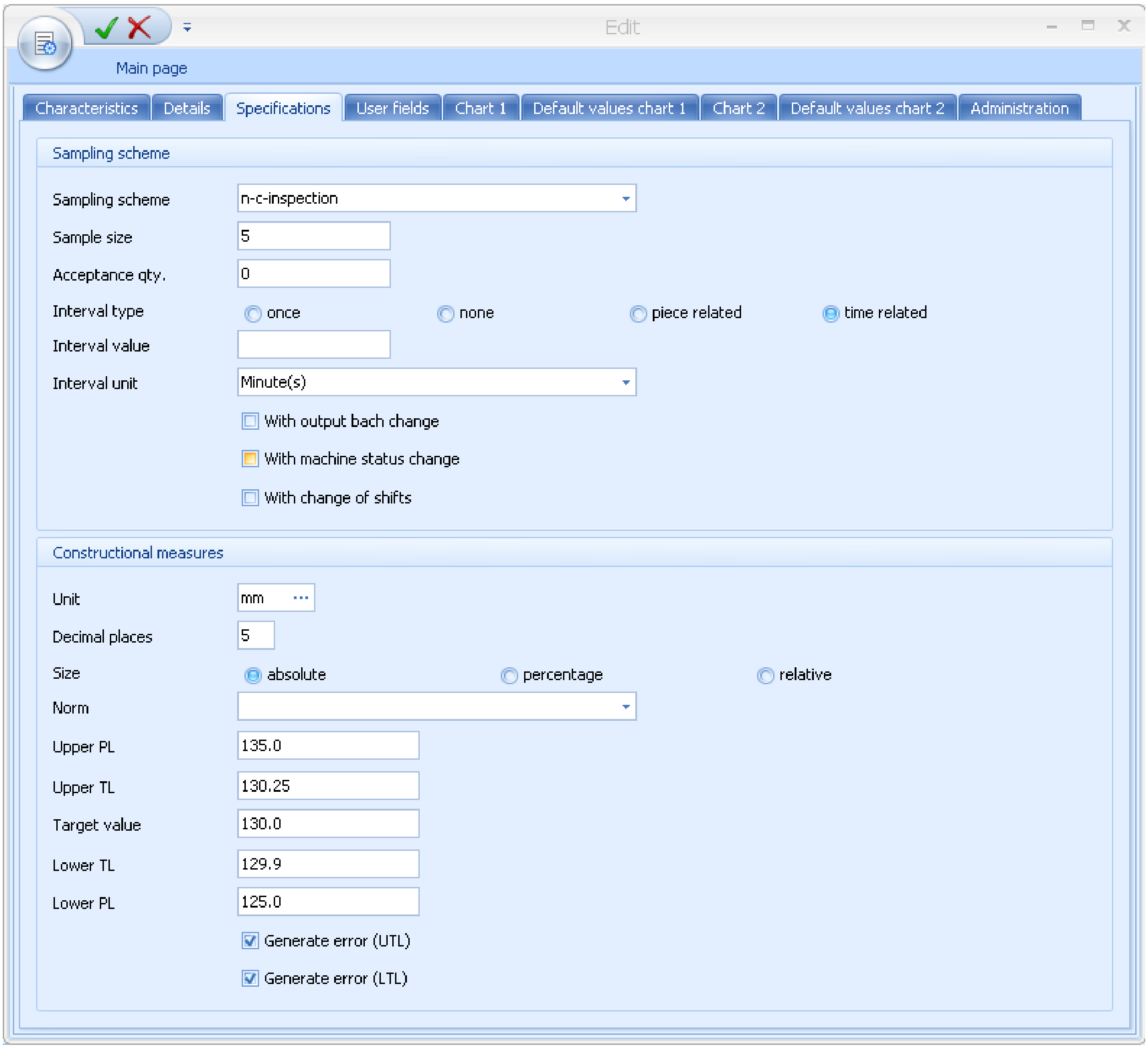
Group *xq*

XQ

Target value, mid-tolerance, mean value of xq chart, input (only visible and can only be selected with an xq control chart)

Editing functions

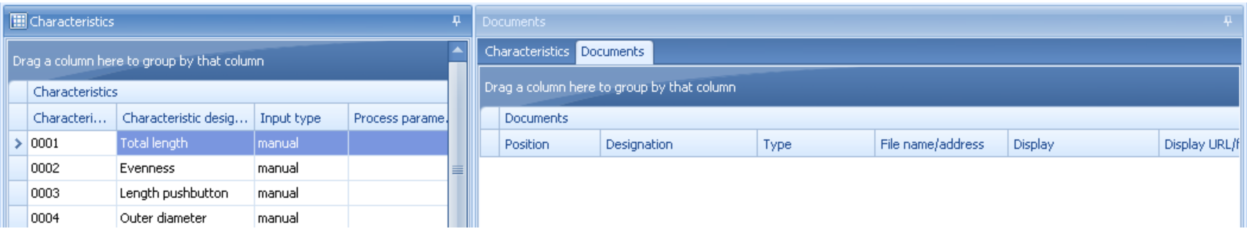
The below screenshot shows an example of an editing dialog. Design and alignment of fields may deviate.



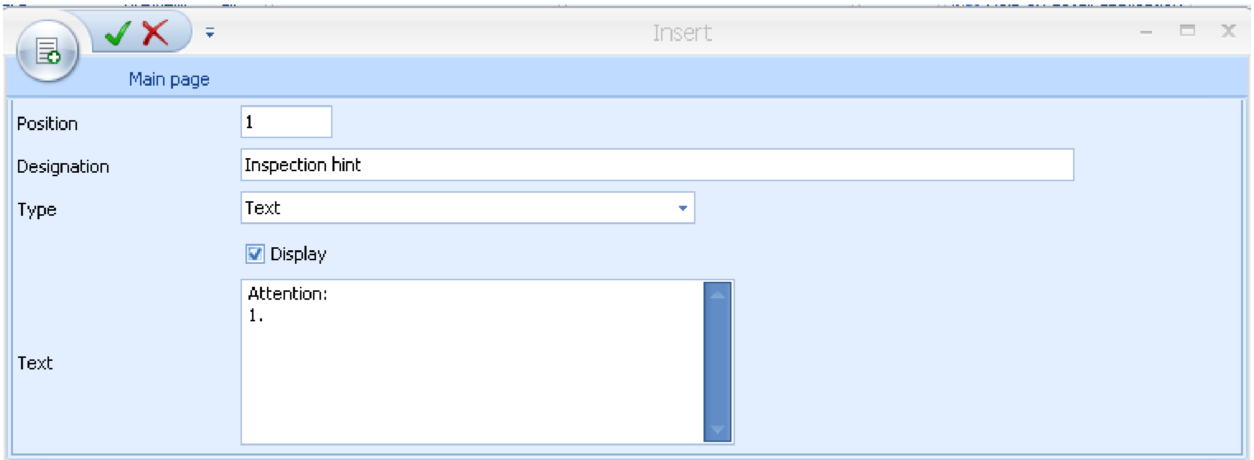
Toolbar

There are no other special function buttons in addition to the standard functions/features.

Detail application *Documents*



If you have activated the tab *Documents*, you can assign an arbitrary number of documents to each characteristic. If this tab is activated, the respective buttons in the toolbar to edit the documents are equally activated.



All formats registered by Windows are available when assigning documents. You can assign simple documents (e.g. written in Word), drawings of any format and videos. You only have to make sure to install a program that is able to display the used format. The appropriate program linked in Windows opens the documents.

The file types "File", "URL", and "Text" are available. If you select the type "file", you can enter the file name including path manually. Select the file type “URL” to access the internet or intranet. Select the file type "text" to directly enter a text.

Note:

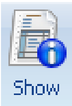
The different types of file format "URL" that the shop floor client supports are listed in the respective manual of the shop floor client. It might happen that "https" URL entries are displayed on the MOC, but not on the AIP shop floor client.

You can assign a designation/name to each document. You can also define the list order of the documents. Use the field "position" to define the order (numeric input). Position numbers must be unique in this list. Enable the checkbox *Display* to define that the document is displayed during inspection process.

Speaking of documents, you also have to decide if a document assignment without precise reference to an article is reasonable. Normally, the document assignment depends on the article.

Taskbar *Document*

In addition to the standard functions, the application also provides the button to show documents.



Show documents

If a document link is stored, click this button to open and show the linked document. However, a program, which can show the linked file type, must be installed on the PC.

1.1 Sampling schemes

The user can select from five sampling schemes in a specified list. Subject to the selected sampling scheme, some additional information has to be defined. It is subject to the subsequent use in the different inspection plan areas (e.g. production, goods receipt, goods issue), if all or only a smaller selection of sampling schemes is available.

Sampling scheme n-c inspection: The sample size is entered in the "sample size" field (= n) and the maximum number of admissible non-conforming units is entered in the "acceptance quantity" (=c) field. The figure "c" is defined as acceptance number. This means: if n = 50 und c = 1, the characteristic and thus the piece is only classified as "fail" if two non-confirming units are identified (with sample size = 50).

Sampling scheme 100% inspection: In general, the sampling scheme *100% inspection* is only used in goods receipt and goods issue. The sample size is calculated from the actual quantity of the inspection requirement and corresponds to it.

Sampling scheme SPC inspection: The sampling scheme "SPC inspection" nearly corresponds to the "n-c" inspection plan. The only difference is that the acceptance limit "c" is not used in this case.

Sampling scheme batch inspection: In the standard configuration, the sampling scheme "batch inspection" only applies to the areas "goods receipt" and "goods issue". The percentage specifying how much percent of the batch is to be checked is entered here. Later in the inspection order characteristic the sample size is calculated from the actual quantity of the inspection requirement and multiplied by the specified percentage.

If you must calculate action limits, you must enter the expected sample size here.

Sampling scheme k-value inspection: With the k value inspection the entered k value is checked against the calculated k value and if this value is violated the sample is rated "fail".

1.2 Control charts for variable characteristics

For variable characteristics, the charts xq, s and R are available.

In statistical quality assurance, production dispersion is used for many calculations. One example is the calculation of capability indices and action limits of a quality control chart. Vice versa, if you have specified a process capability index, you can estimate the production dispersion and calculate the action limits on this basis.

The specifications for the calculation of limit values can be found in the tab "default values chart 1" or "default values chart 2", where values to estimate the production dispersion can be entered. The action and warning limits can be calculated on the basis of these specifications. However, it is also possible to enter the production dispersion directly. The system provides three calculation options.

You first describe the specifications using the xq and s chart. The differences with the R chart are explained in more detail in the sections that follow.

There is often a specification for the process capability index cpk. This specification is reasonable. If the process capability index cpk is respected, you can then produce pieces within the range of tolerance. Based on the specified cpk value, the system calculates internally an estimated value for Sigma, which is entered to the right of the option "Sigma" for information purposes. The estimated basic value that has been calculated is used to calculate the limit values of the xq/s chart. The calculation is performed, once further data has been entered using the *Calculate* button. The calculation method "cpk" is set by default. In addition, there are also the calculation methods "sigma" and "sq/an".

The cpk value of 1,33 ensures that 99.725% of the characteristic values are within the tolerance. However, it is often required that 99.994% of the characteristic values are within the tolerance limit, which corresponds to a cpk value of 1.67.

The calculation method sq/a_n means that an estimate of the standard deviation is calculated from the quotient of the medium standard deviation and a correction factor a_n. This correction factor depends on the sample size, which is identified by the index n. The values for a_n are defined in the system and are requested automatically. This estimate of the standard deviation is best in case that there is no specification of the process capability index and the production dispersion is unknown and thus the specification of the sq-value has still to be corrected by a correction factor. It is also the most efficient method under the given conditions. You must specify the sq-value to calculate the limit values later on. Enter the value in the field on the right hand side of the option sq/a_n. If you click the button *Calculate* later on, the estimate sq/a_n is calculated using the specified sq value. The result is entered on the right hand side of the option Sigma for information purposes. This estimate is then the basis for the calculation of action and warning limits of the xq/s chart

The third calculation method requires the specification of a sigma value. In this case it is assumed that sigma is known and consequently the correction factor is not required. Enter the Sigma value to the right of the “sigma” option. In comparison to the previous method, sq/a_n is replaced by sigma. In the majority of cases sigma is not known. Therefore, it is best to use the calculation method using the specified sq value to automatically calculate the estimate sq/a_n for variances in case of doubt.

If you select the “relative deviation from xq” or the “deviation from xq in percent” as “specification to calculate limit values”, the input option for “action probability in %” disappears. Instead, you can enter the “deviation from target value”. These values and the specified value of xq are then used to calculate the limit values (target value, middle of tolerance, mean value of xbar chart, input).

Further details have to be made in order to identify action and warning limits of the xq chart. You must specify an xq value. The system offers the possibility of setting the xq value equal to the middle of tolerance or the target value or of specifying a value manually. If the process is supposed to be aligned to the mean value, the middle of tolerance should be preferred as xq value.

The action probability must be entered in percent in order to calculate action and warning limits of the xq/s-chart. For this purpose, you must first dedice, if you want to use one-sided or two-sided limit values for the calculation. Selct one of the two options.

Once you have specified the option 'one-sided' or 'two-sided', enter the action probability in percent. The possible and reasonable action probabilities are defined in the system and only need to be selected from the list. For the xq-chart, the calculation is based on the standard distribution. For example, if 99.725% of the characteristic values must be within the action limits, select the value 99.725. As the specification of a sigma area is commonly used by some users, the corresponding sigma area is displayed with the respective action probability for information purposes.

If warning limits are also to be calculated, an action probability must be entered here. Note: The probability value of the warning limit must be lower than the action limit value.

The sigma area is not displayed in the selection list, since the distribution of chi² is used to calculate the limit values of the s-chart. Apart from that, the input is the same as for the xq chart.

If you save the specifications, the limits are calculated – if the *Calculate* checkbox has been enabled before.

As already mentioned, the user can enter the limit values directly without any specified values.

If the R-chart is selected instead of the xq or s-chart, the option sq/a_n is replaced by Rq/d_n in the specifications of the calculation. The estimate of sigma is now calculated using the specified mean range R divided by the correction factor d_n. This correction factor depends on the sample size n. The corresponding values are defined in the system and are selected automatically. Apart from that, the rest is the same as for the xq or s-chart. Note: In case of an R-chart, the calculation of the limit values is not based on a chi² distribution, but on a table stored in the system, which is based on standardized ranges.

Notes:

- You can only use the calculation specifications "relative/percentage deviation from xq" if you use an xq control chart. For this reason, the fields of the group "Xq" are only visible, if you have previously selected the xq control chart.
- The user must select the confidence interval (one-sided/two-sided). Usually, you do not define a lower limit for an s-chart. In this case, select "one-sided".
- If only one of the two tolerance limits is available, you cannot use the calculation method "cpk". The calculation formula of the cpk method requires both tolerance limits.

1.3 Control charts for attributive characteristics

The p- and u-charts are available for attributive characteristics.

p identifies the proportion of defective units in the sample and u identifies the failures/defects per unit in the sample. As to the p chart, it is important that each item is either defined as defect-free or defective. If an item has several failures/defects it is only once referred to as defective.

In contrast to the variable characteristics, there are no lower limit values. Furthermore, it is normally not necessary to state the values UTL, LTL and target value.

It is necessary to enter a pq or uq value in percent for the automated calculation of specifications. This can be done in the default values tab.

If you save the specifications, the limits are calculated – if the *Calculate* checkbox has been enabled before.

Calculation is respectively based on normal distribution. The value 99,725 has to be selected if, e.g. 99,725% of the characteristic values are supposed to lie below the upper action limit. As the specification of a sigma area is commonly used by some users, the corresponding sigma area is displayed with the respective action probability for information purposes.

1.4 Calculation of formulas

If you store a formula, you can automatically calculate measured values by way of measured values or statistical values of other characteristics that have been inspected before.

If the extension QMSingleValue.FormulaArguments is enabled, you have the possibility to use extensive arguments to calculate the single value you want to collect. In addition, you have more possibilities to access specification values and values of inspection results of other characteristics. For more details, refer to the section "".

If this extension is not enabled, you can only calculate characteristics using the inspection results of other characteristics that have already been entered. Find details in the section "Calculation via reference to other inspection results".

1.4.1 Operators, functions and constants

The following operators, functions and constants for calculating measured values are supported:

Functions	
abs(x)	Calculates the absolute value
atan(x)	Calculates the arc tangent
cosh(x)	Calculates the hyperbolic cosine
float(x)	Converts the value into a floating point number
sqrt(x)	Calculates the square root
acos(x)	Calculates the arc cosine
atan2(y,x)	Calculates the arc tangent of y/x
exp(x)	Calculates the exponential value
log(x)	Calculates the natural logarithm
sin(x)	Calculates the sine
tan(x)	Calculates the tangent
asin(x)	Calculates the arc sine
cos(x)	Calculates the cosine
int(x)	Converts the value into an integer
log10(x)	Calculates the common logarithm
round(x)	Rounds to integer value
round(x,y)	Rounds the value x to y decimal places
sinh(x)	Calculates the hyperbolic sine
tanh(x)	Calculates the hyperbolic tangent
trunc(x)	Reduces the value x to an integer value
trunc(x,y)	Reduces the value x to y decimal places
Operators	
x + y	Addition
x – y	Subtraction
x / y	Division
x * y	Multiplication
x ** y	Calculates x to the power of y
Constants	
pi	3.141592654
e	2.718281828

If constant numeric values are used in formulas, you must be careful not to use thousand separators. If these constants are floating point numbers, be careful to use a dot as decimal separator instead of a comma.

1.4.2 Formulas referring to other inspection results

Formulas including a reference to other inspection results are always calculated when an inspection result referenced in the formula is created, changed or deleted.

For these characteristics, you must first specify the level of the formula calculation. The following types are available:

- V – Calculation on the level of single values (Value).
  For each single value of the characteristics involved, one single value is generated for the calculated characteristic.

- S - Calculation on the level of samples (Sample).
  For each sample of the characteristics involved, exactly one single value is generated for the calculated characteristic.

- C - Calculation on the level of characteristics (Criteria).
  Exactly one single value is generated for the calculated characteristic (with respect to the overall statistic of all characteristics involved)

The actual formula follows this identifier (see previous chapter).

The following syntax applies for the variables identifying the single values or statistical values of the order characteristics involved [x:y:z].
The x parameter identifies the statistical value to be used. The available values are listed below. Please bear in mind that the calculation level might cause restrictions.

- `X` – Single value
  (is only available for calculations on the level of single values)

- `AVG` – Mean value
  (is only available for calculations on the level of samples or characteristics)

- `MIN` – Minimum
  (is only available for calculations on the level of samples or characteristics)

- `MAX` – Maximum
  (is only available for calculations on the level of samples or characteristics)

- SUMX – Sum of single values

  (is only available for calculations on the level of samples or characteristics)

- R – Range

  (is only available for calculations on the level of samples or characteristics)

- S – Standard deviation

  (is only available for calculations on the level of samples or characteristics)

- N – Sample size

  (is only available for calculations on the level of samples or characteristics)

- M – Number of samples

  (is only available for calculations on the level of characteristics)

The y parameter describes how the corresponding characteristic is supposed to be identified. The following possibilities are available:

- SENO – identification via the OP sequence of the characteristic (serial number)

- INCR – Identification via the characteristic number (inspection criteria)

  If the characteristic number is not unique within the inspection requirement, it is not predictable which one of the applicable characteristics is used at the time of calculation.

The parameter z identifies the characteristic using the field content defined by parameter y. Either the OP sequence or the characteristic number of the calculation source is entered in this field. If the characteristic number includes a space character, it should be replaced by an underscore within the formula.

Example 1:

A new characteristic is calculated from the single values of the characteristic assigned to the number "LENGTH"/"LAENGE" divided by 2.5. A corresponding single value is supposed to be calculated for each single value of the source characteristic (calculation on the level of single values).

→ Formula: `V: [X:INCR:LAENGE] / 2.5`

Example 2:

The characteristic "surface" results from the product of the characteristics with the characteristic number "LENGTH"/"LAENGE" and "WIDTH_TOTAL"/"BREITE_GES". A single value of the characteristic "surface" is supposed to be calculated for each single value of both source characteristics (calculation on the level of single values).

→ Formula: `V: [X:INCR:LAENGE] * [X:INCR:BREITE_GES]`

Example 3:

The characteristic "maximum margin width" results from the subtraction of the minimum of the characteristic "inside diameter" (OP sequence 10) from the maximum of the characteristic "outside diameter" (OP sequence 20). A single value of the characteristic "maximum margin width" is supposed to be calculated for each sample of both source characteristics (calculation on the level of samples).

→ Formula: `S: [MAX:SENO:20] - [MIN:SENO:10]`

For the calculation of formulas including references to other inspection results, it is allowed to calculate new formula characteristics that are based on calculated formula characteristics. However, this nesting may not have more than 10 references one below the other. Furthermore, double concatenations must not be created (Example: characteristic A is calculated from characteristic B and characteristic C; characteristic C is calculated from characteristic A).

1.4.3 Extended formulas

Extended formulas are only available, if the extension QMSingleValue.FormulaArguments is enabled.

The extended formulas provide the following advantages compared to the formulas including references to other inspection results:

- You can enter arguments for these characteristics that are used to calculate the measured value. In most cases, you do not need to use other "source characteristics".
- On saving the inspection result, the measured value is calculated and is immediately available. You do not need to refresh the measured values in the AIP to see the measured values.
- For the calculation, you can optionally use single values or sample or characteristic statistics of other characteristics. You may combine these in any way.
- You can use variables for the target value, the upper and the lower tolerance limit of the current characteristic or of other characteristics in the formula.

For this reason, you should primarily use the extended formulas.

The following syntax applies for the variables identifying the single values, statistical or specification values of the order characteristics involved `[x:y:z]`.

The x parameter identifies the statistical value to be used. The available values are listed below. Note: Depending on the respective shop floor client used, it is possible that not all 10 argument fields are available.

- `VAR1` – Argument 1 of the inspection result of the own inspection step characteristic
- `VAR2` – Argument 2 of the inspection result of the own inspection step characteristic
- `VAR3` – Argument 3 of the inspection result of the own inspection step characteristic
- `VAR4` – Argument 4 of the inspection result of the own inspection step characteristic
- `VAR5` – Argument 5 of the inspection result of the own inspection step characteristic
- `VAR6` – Argument 6 of the inspection result of the own inspection step characteristic
- `VAR7` – Argument 7 of the inspection result of the own inspection step characteristic
- `VAR8` – Argument 8 of the inspection result of the own inspection step characteristic
- `VAR9` – Argument 9 of the inspection result of the own inspection step characteristic
- `VAR10` – Argument 10 of the inspection result of the own inspection step characteristic
- `X` – Single value of another characteristic
- `AVG` – Mean value of the sample of another characteristic
- `MIN` – Minimum of the sample of another characteristic
- `MIN` – Maximum of the sample of another characteristic
- `SUMX` – Sum of the single values of the sample of another characteristic
- `R` – Range of the sample of another characteristic

- `S` – Standard deviation of the sample of another characteristic
- `SREL` – Relative standard deviation of the sample of another characteristic
- `N` – Sample size of another characteristic
- `AVG_ALL` – Mean value of all samples of another inspection step characteristic
- `MIN_ALL` – Minimum of all samples of another inspection step characteristic
- `MAX_ALL` – Maximum of all samples of another inspection step characteristic
- `SUMX_ALL` – Sum of the single values of all samples of another inspection step characteristic
- `R_ALL` – Range of all samples of another inspection step characteristic
- `S_ALL` – Standard deviation of all samples of another inspection step characteristic
- `N_ALL` – Total sample size of all samples of another inspection step characteristic
- `M_ALL` – Number of samples of another inspection step characteristic
- `TV` – Target value of an inspection step characteristic
- `UTL` – Upper tolerance limit of an inspection step characteristic
- `LTL` – Lower tolerance limit of an inspection step characteristic

The `y` parameter describes how the corresponding characteristic is supposed to be identified. The following possibilities are available:

- `SENO` – identification via the OP sequence of the characteristic (serial number)
- `INCR` – Identification via the characteristic number (inspection criteria)
  If the characteristic number is not unique within the inspection requirement, it is not predictable which one of the applicable characteristics is used at the time of calculation.

The characteristic number must not include any special characters. A minus sign "-" is not permitted, for example.

- SELF – Identification of the own calculated characteristic
  The characteristic that is to be calculated identifies itself. Only in this case, the parameter z is not required.
  Note: You may only use the identification of the own characteristic for the argument fields and for the target value and the tolerance limits.

The parameter z identifies the characteristic using the field content defined by parameter y. Either the OP sequence or the characteristic number of the calculation source is entered in this field. If the characteristic number includes a space character, it should be replaced by an underscore within the formula.

Example 1:

The measured value of the current characteristic is calculated from the sum of the argument fields 1 to 4.

→ Formula: `[VAR1:SELF] + [VAR2:SELF] + [VAR3:SELF] + [VAR4:SELF]`

Example 2:

The characteristic is the result of the product of the maximum measurements of the inspection step characteristics with the characteristic numbers 'LAENGE' and 'BREITE_GES' ('LENGTH' and 'WIDTH_TOTAL').

→ Formula: `[MAX_ALL:INCR:LAENGE] * [MAX_ALL:INCR:BREITE_GES]`

Example 3:

The measured value is calculated from the sum of the following three summands:

- Content of argument field 1
- Middle of the tolerance of the current characteristic
- Sample mean value of the characteristic with OP sequence 10

→ Formula: `[VAR1:SELF] + (([UTL:SELF] + [LTL:SELF]) / 2) + [AVG:SENO:10]`

Note the following when using extended formulas:

- Contrary to the formulas including references to other inspection results, the measured values are not calculated when the "source characteristics" are changed. The measured values are only calculated, if the inspection result of the respective calculated characteristic is explicitly collected or changed (e.g. via the argument fields).
- When the inspection result is saved, the system must be able to identify valid values for all variables used in the formula (single values, sample or characteristic statistics, specification values of other characteristics, all used arguments).
  Otherwise, an error message occurs and the inspection result is not saved.
- You cannot directly edit the calculated measured value. The measured value is always the result of a calculation.
- If you use the parameter `[X:…]`, the respective single values of other characteristics are searched for using the absolute single value and sample number. For the current characteristic, the parameter `[X:…]` is not available.
- If you use the statistical parameters `[MAX:...]`, `[MIN:...]`, `[AVG:...]`, `[SUMX:...]`, `[R:...]`, `[S:...]`, `[SREL:...]` or `[N:...]`, the respective statistical values are searched for using the absolute sample number. Here, you cannot use statistical parameters of the own characteristic.
- If you want to use the statistical parameters of the complete characteristic using the parameters `[MAX_ALL:...]`, `[MIN_ALL:...]`, `[AVG_ALL:...]`, `[SUMX_ALL:...]`, `[R_ALL:...]`, `[S_ALL:...]`, `[M_ALL:...]` or `[N_ALL:...]`, only the data of other characteristics is available (not the data of the own characteristic).
- Via customization, extensions can be made available to obtain any variables in the syntax `[VAR:`***`<Object>`***`:`***`<Identifier>`***`]`.
- You cannot use characteristics that include extended formulas as sources to calculate formulas including references to other inspection results. But you can use these characteristics for other characteristics with extended formulas.

1.4.4 General notes on calculated characteristics

If unknown variables are used within a formula (faulty parameters `x` and/or `y`), the escalation `CPAUMW.CALCULATED_CRITERIAS_GET_VARIABLE_VALUE` is triggered.
If problems occur on assigning an identified value to a variable of the formula, the escalation `CPAUMW.CALCULATED_CRITERIAS_SET_VARIABLE` is triggered.
Both actions described require the escalation management license.

Tool numbers, machine numbers, cavity numbers or similar information are not stored for the calculated single values.

To transfer a corresponding number (batch number, sample number, serial number, etc.), all source samples of the calculation must be assigned the same number. If there is no number that is assigned to all source samples, you cannot assign a number to the calculated sample. If several numbers are found that have been assigned to all source samples, only the first number found is assigned to the calculated sample.
This function only applies to numbers, which have been assigned on sample level.

1.5 Last off inspection

As part of the function extension for the in-production inspection, the function of the last off inspection is available. For this function, you must have created the CAQ system option 1222 manually as a precondition. For details, refer to the procedure document "Configuration_QM_Options.pdf". The documentation of the CAQ system option specifies which characteristic user fields must be created (master data characteristic, inspection plan characteristic and inspection step characteristic), so that you can specify a characteristic for a last off inspection.

To specify a characteristic for a last off inspection, the user field of the last off inspection must have a content.

The function "Last off inspection" is not offline capable. And you cannot use the last off inspection with operations that are specified as *Inspection OP* via processing code.

If the operation is logged off or interrupted in offline mode, the buffered activities/postings are processed one after the other when the online mode is restored. This has the effect that the operation is logged off or interrupted although the last off inspection is missing.

The processing code generally defines if any check for a defined last off inspection is performed at all during logoff/interruption. If the processing code in tab *Quality* is set to *Inspection OP*, no check and no last off inspection is performed.

If an inspection step has been "logged on" with the logon of an operation on the AIP, the system proceeds as follows for the last off inspection when the operation is logged off or interrupted.

1. The system checks if an inspection point with cause for creation *Last off inspection* exists for this inspection step at the workplace in question. It does not matter if the inspection point is completed or not. If the check is also performed with an interruption, the system checks if an inspection point with cause of creation *Last off inspection* has been created since the last logon. If an inspection point is found, the operation is interrupted or logged off.
2. If no inspection point is found in item 1, the system checks if the relevant inspection step logged on includes characteristics with the inspection due date *Last off inspection*. If this is not the case, the operation is logged off or interrupted.

3. If the processes described in item 1 and 2 have the result that an inspection point must be created with the cause of creation *Last off inspection*, the following message is shown:

   Last off inspection is missing!
   Enforce posting using the option "posting required"?

   Using the option *Posting required*, you can perform the logoff/interruption without having made a last off inspection. Use the CAQ system option 1154 to activate the logoff/interruption using the option *Posting required.*

4. If the operation is not logged off or interrupted because of the missing inspection point with the cause of creation *Last off inspection*, you must go to the inspection to create an inspection point with the cause of creation "Last off inspection". You can use the option *Last off inspection* in the inspection list on the level *Inspection step* to create an inspection point with cause of creation *Last off inspection*.

If you try to log off or interrupt the operation that includes an inpsection point with cause of creation *Last off inspection* that has not been completed, the standard processes apply. This means that the operation can optionally be logged off or interrupted although the last off inspection has not been performed/completed using the option "Posting required".